

Recommendation Report for Copper-Nicus Mining Ltd. Underground Mine Development

1. Scope of Work Completed

In response to a request from Copper-nicus Mining Ltd. looking into copper mining sites in the Nicus Mountain Range of Northern British Columbia, J-8 has performed an evaluation of three potential copper mines. We have collaborated with stakeholders to analyze and rank the three deposits in terms of social, economic, and environmental impacts to determine the best deposit for further development.

Key Deliverables:

- Letter of Transmittal
- Recommendation Report with recommended mining site to develop.
- Oral Presentation

Our recommendation is constrained by the lack of access to extra resources on the Nicus Mountain Range, local communities/First Nations, and stakeholder input. Thus, the recommendation report is based on the assumption that available pricing/cost data remains constant throughout the project duration and that the block models we have been provided with are accurate.

2. Engagement, Learning, and Synthesis Strategies

The community, including Indigenous rights holders and residents of the nearby town, will be invited to a number of public meetings, consultations, and polls during the entire process. They will be invited to give their opinion and share perspectives throughout all stages of the process.

Stakeholder Group	Engagement Goal	Engagement Strategy (Timing and Format)	Benefit to Project/ What is expected to be gained
General Community	Inform	Public meetings and news, throughout the entire process. Online discussion forums.	Gain opinions, and the social licence to operate of people living near the site.
Indigenous Peoples	Collaborate/ Involve	Consultation meetings and polls, before significant decisions are made. Discussion forums. Progress reports and input opportunities throughout the mine life cycle.	Gain trust and a social licence to operate on traditional land.
BC Government	Collaborate	Formal reports and discussions at checkpoints throughout the entire process. (ie. approval of preliminary plan)	Legal funding and approval.
Mining Company	Collaborate	Formal reports and discussions throughout the entire process.	Approval, and suggestions on mining.

The combination of engagement processes shown in the table above are the most efficient and provide the greatest benefit to the project, as engagement with all key stakeholders happens consistently, yet does not require a large cost to perform. Engagement with large stakeholder groups focuses on online

engagement activities that require less effort and cost spent, while still conveying information and obtaining the required feedback. Conversely, primary stakeholders are provided with formal reports and discussions as their input is essential.

Project success can thus be measured not only in terms of technical achievement but also community acceptance. This can be done by taking data from polls and discussions and determining whether final decisions are in alignment with the interests of stakeholders. If any unforeseen difficulties arise, there will be extra costs in the initial budget laid out to increase community engagement, obtain more feedback, and revise important decisions if necessary. We hope that with a strong community engagement strategy, the social licence to operate remains strong.

3. Analysis of Mine Sites

3.1 Deposit 1

3.1.1 Description of the Deposit

Deposit 1 is a deep, narrow, and remote deposit that contains a high grade of copper ore (see Appendix B). It is located in a remote, mountainous location, to the west 300 km away from the main highway and local community, and 150 km away from the Nicus River (See Appendix F). The deposit sits 300 m below the surface, making underground mining the only economically viable approach, as open pit mining would require an exorbitant amount of waste rock, causing it to be unprofitable. Underground mining also produces significantly less tailings, which, combined with the high-grade ore, mean less treatment and thus less tailings storage required. This minimises environmental impact. Furthermore, the deposit's remote location minimises disruption to existing communities and land with archeological and traditional significance.

The primary challenge for this deposit is its high development cost. With no existing infrastructure in close proximity, construction of new roads and powerlines must be built across the Nicus River and through the rugged, mountainous terrain to reach the deposit.

3.1.2 Environmental Impact of the solution

The environmental impact for the development and use of the mine itself isn't that severe; high-quality copper ore requires less water to process and extract copper concentrate, leaving less tailings. The waste rock, 34,965,000 tonnes, is on the lower end when compared to Deposit 3, for example (see Appendix B). However, this assessment does not account for the construction of new infrastructure, which is likely to be more environmentally disruptive than the mine itself.

Building roads and powerlines across the 300 km of rugged, mountainous terrain will require clearing of forests and vegetation. A bridge would need to be constructed over the Nicus River, requiring extra erosion and sediment runoff management, as it will affect fish stocks and water quality in the lake reserve below. Furthermore, power transmission corridors will require further deforestation on top of the road clearing. Overall, this makes the cumulative environmental footprint for Deposit 1 extremely significant.

3.1.3 Economic Viability and Sustainability

Deposit 1 provides a total profit of \$6,464,137,500 from mining alone; this makes it economically competitive with Deposits 2 and 3. However, this doesn't account for the significant infrastructure costs required to reach the deposit. The construction of 300 km of road, a bridge over the Nicus River, and power transmission systems will consume a significant proportion of the mining profit. A rough

cost estimate suggests a loss of 33-59% of profits from the mine (see Appendix E), which depends a lot on the terrain difficulty and construction requirements. Of all three deposits, Deposit 1 carries the highest infrastructure cost and uncertainty, as a result, is expected to have the longest break-even period.

3.1.4 Social Sustainability

The social sustainability of Deposit 1 is defined by its isolation, which acts as both a strength and a weakness. Its location 300 km away from the local community and main highway ensures that the project avoids the negative social impacts common to mining, such as noise, dust, and visual pollution, to residential areas. This distance also allows for the preservation of land with archaeological and traditional significance. However, this same remoteness limits the project's ability to foster community engagement and provide local benefits. Unlike the sites closer to town, Deposit 1 offers fewer opportunities for local employment or the construction of shared infrastructure. This lack of integration is reflected in the community benefit score of 0.8, which is the lowest among the three deposits. While the site avoids direct opposition, it fails to establish the connection with the local communities.

3.1.5 Other considerations

A consideration for Deposit 1 involves the potential for a second mineshaft being added to the underground mine. While a single shaft is the baseline for the proposed underground mine, another mine shaft would make the mine much more resilient by improving safety for workers in the deep and narrow deposit. However, the cost of adding a second mine shaft is approximately 600 million dollars. This represents a significant additional cost that must be evaluated against the total projected profit. Given the deposit already requires the construction of new roads and power systems infrastructure through 300 km of mountainous terrain and the Nicus River, the high cost of a second shaft could impact the overall feasibility of the project.

3.2 Deposit 2

3.2.1 Description of the Deposit

Deposit 2 is a moderately deep, moderately remote deposit with a large lateral extent. Underground mining is the best approach as ore grade increases with depth. Additionally, this deposit is located within 50 km of the Nicus River that flows into Nicus Lake (see Appendix F), which acts as a food source to local communities. This means reducing runoff from tailings storage is a high priority and further justifies using an underground mine, as an open pit mine would require 8.5x the waste rock to maintain profit margins. A potential mine at Deposit 2 would benefit the nearby community economically and with new infrastructure. However, infrastructure, underground mining, and environmental impact mitigation all add to the high cost required to mine here.

3.2.2 Environmental Impact of the solution

As stated in the deposit description, reducing runoff is a high priority for this deposit. Therefore, 40% backfill plus dewatering will be used to reduce the required volume of the tailings dam. To meet government and community standards, and without impacting fishing in a nearby lake, water containment and filtration must be added. This has an expected cost of around \$50 million per year after closing, which can be covered by interest on a \$960 million support fund (see Appendix C, Deposit 2). Another benefit of Deposit 2 is that the underground mine will significantly reduce the cost of remediating the land above the mine.

3.2.3 Economic Viability and Sustainability

This mine will bring a net potential profit of \$7.1 B to be shared between Copper-Nicus Mining Ltd. and the represented First Nations, with \$100M set aside for a new community centre, high-speed wifi, and an ice rink, and \$960M set aside in the environmental support fund. For the First Nations, the proximity of this mine to the local community and the support fund will bring a long-lasting source of jobs and cash inflow.

This deposit will require 1.5 years to build infrastructure and has a break-even period of 7-14 years. This mine also has lower certainty in the exact deposit size and ore quality than others; however, the high profit margin should improve the resiliency in the event they are below the expected amount.

3.2.4 Social Sustainability

Creating a mine at Deposit 2 allows significant community involvement throughout the mine's life cycle, as community and First Nation members will be involved in both mine planning/development as well as managing the support fund after closure. Its location avoids important archeological sites while also ensuring that developed infrastructure (power lines, bridge, access road) will remain close to inhabited areas along roads, the Nicus River, and Nicus Lake, as requested by the First Nations. Additionally, despite being the closest mine to the Nicus River and Lake, appropriate funding has been allocated to prevent negative impacts on the water, preventing pushback from local residents, fishing groups, and the BC Government.

3.2.5 Other considerations

The environmental support fund of \$960M can also be used to replace the existing water containment and filtration system at the tailings dam in the event that the represented First Nations find its performance unsatisfactory or future technology better supports the tailings impoundment area.

3.3 Deposit 3

3.3.1 Description of the Deposit

Deposit 3 is a large, shallow and broad deposit with lower grade ore that would be mined as an open pit. The location near a town makes infrastructure and operation easy, and since this deposit is shallow, there is not too much waste rock in proportion to ore which lowers the environmental impact along with making the site very resilient. The challenge lies in tailing storage, as the deposit is located very close to archaeological sites and land with cultural significance, yielding strong opposition from the community.

3.3.2 Environmental Impact of the solution

Developing Deposit 3 into an open pit mine has large environmental impacts, especially during the closure phase. During each stage considered in the SCLAs (see Appendix C for Deposit 3), the emissions and waste generation for Deposit 3 score ratings of 1 or 2. This is on a scale of 1 to 5, where 1 is the lowest score. Due to its location beside a town, it is critical that the emissions and waste are properly managed. This includes having a detailed closure plan and waste management strategies for tailings. To ensure the criteria outlined by the stakeholders and rights holders is met during the entire life cycle of the mine, care would need to be taken to prevent contamination to the nearby town and archeological site. The archaeological site has cultural significance to the local Indigenous people, and there is concern about storing tailings as a slurry nearby. Due to the importance of the land, dry stacking the tailings would be the best option to ease worries of contamination. Upon closure, land reshaping of the large pit and stabilization of the walls would be used to help restore the landform.

3.3.3 Economic Viability and Sustainability

For Copper-nicus, Deposit 3 has good economic viability. Using the block model simulations (see Appendix B), we found that this deposit would generate a raw profit of \$2.67 billion using open pit mining. In addition, since Deposit 3 is directly beside a town, less additional infrastructure would need to be built, resulting in a lower pre-mining cost. The large size of the deposit and the proximity of the ore to the surface make it cost-effective compared to underground mining. There is an anticipated break-even period of 5–8 years. The economic impact on the stakeholders and rights holders would also be positive. A large mine increases the job opportunities and cash flow to the town, resulting in greater spending money for public town projects. We understand that Copper-nicus Mining has identified a new community center and ice rink as items of interest to the local community. Setting aside a portion of the profit to help fund the development of these two buildings would increase the agreement among stakeholders and strengthen the project's social licence to operate.

3.3.4 Social Sustainability

Deposit 3 is located by a nearby town, which means that it is close to the local community. The development of this mine would lead to the creation of more training or work opportunities for the population, resulting in economic stability, hence benefiting both the local population and businesses. Copper-Nicus Mining Ltd. has also proposed various initiatives that could benefit the local community. These benefits include the construction of a new community centre and ice rink, improvement in the infrastructure of roads, and high-speed internet access. These initiatives would strengthen the relationships between the community and the company, building a strong social license to operate.

However, there are several social issues surrounding the mine's location near the archaeological site, as this land is of cultural significance to the local First Nations. Mining in this area could result in opposition from the First Nations people. It is therefore crucial to have consistent consultation with the First Nations rights holders in order to achieve social sustainability. It is also important that Copper-Nicus Mining Ltd. work hand in hand with the First Nations people through various programs, such as environmental monitoring or revenue-sharing agreements, in order to build trust and ensure that the interests of the First Nations people are being considered and respected.

3.3.5 Other considerations

There are several other factors which should be considered when analyzing the practicality of developing Deposit 3, one being the approval process to mine in an area near an archaeological site with cultural significance to local First Nation communities. This process would consist of several additional environmental assessments or government regulations, which would make the process more tedious or lead to resistance from the local communities if not handled correctly. Another aspect that must be considered would be the tailings pond storage facility. These facilities store water behind dams; however, if the dam were to break, this would cause extreme environmental damage. It would be beneficial to use a dry stack method instead or implement drainage systems, as this would remove the water from the mining material, preventing any contaminants from entering the surrounding soil or groundwater. A final consideration would be extreme weather factors such as heavy rainfall, which may put a pause on any mining activities, so it is crucial that proper planning is done to prevent this.

4. Final Selection and Justification

Our three SLCA's (see Appendix C) used Health and Safety, Water Use, Waste Generation, and Emissions as the four criteria to evaluate the sustainability during development, operation, and closure of all three mine deposits. Deposit 2 scored the highest value of 28 out of a possible 36 due to its low waste generation, emissions, and water use. Deposit 1 has a similar score of 24 but is less desirable due to extremely low scores surrounding health and safety for operating such a deep and remote mine. Deposit 3 scores the worst through SLCA analysis, as large amounts of waste rock are produced, requiring extra water use throughout its entire life cycle.

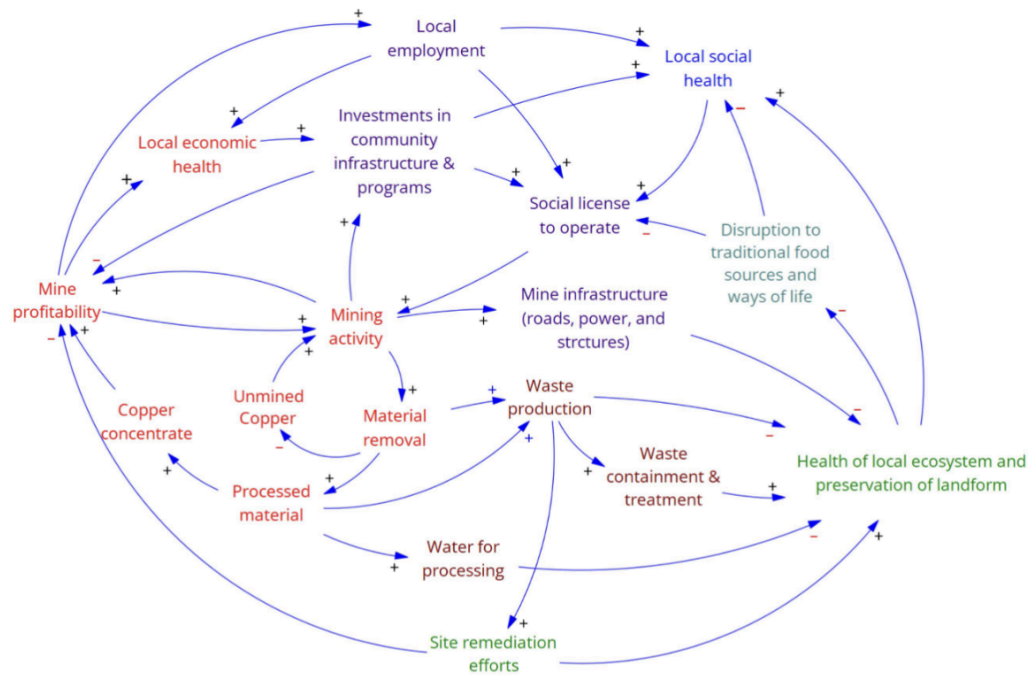
The four criteria and their scores from the SLCA's were then added to the WDM by averaging the results across the three life cycle stages (see Appendix D) and assigned a collective 45% weighting (25% for Lifetime Health and Safety, 10% for Lifetime Waste Generation, and 5% for emissions and water use), representing the sustainability of the deposits. The other criteria considered are the total profit from the mine, preservation of culturally significant land, and benefit to the community. Profit is weighted the most at 30% as this is a need with high stakeholder alignment. Through our final WDM analysis, Deposit 2 has the highest score of 6.82, which is a lot higher compared to a score of 6.43 for Deposit 1 and 5.27 for Deposit 3. Sensitivity analysis for robustness of data consistently shows that Deposit 2 is the ideal option.

An underground mining approach best preserves the landform and has the least environmental impact as long as the required safety precautions and proper tailings storage are implemented. From an economic standpoint, Deposit 2 will have a resulting profit of \$7.1B after the creation of new infrastructure, a community ice rink and community center, and a \$970M environmental support fund. It will provide jobs and proper training throughout the mine's life cycle and for a long period of time afterward for tailings impoundment area maintenance. Socially, this deposit is also the best one as it does not impact any archaeological/traditional lands and does not disrupt the way of life of Indigenous communities along the Nicus River. Consistent engagement with stakeholders (as detailed in Section 2) will also help the project retain a strong licence to operate. Ultimately, Deposit 2 is the most viable option throughout all the dimensions of sustainability as it produces positive impacts to society, the environment, and the economy.

Our recommendation for an underground mine at Deposit 2 best balances the needs of all stakeholders with the dimensions of sustainability.

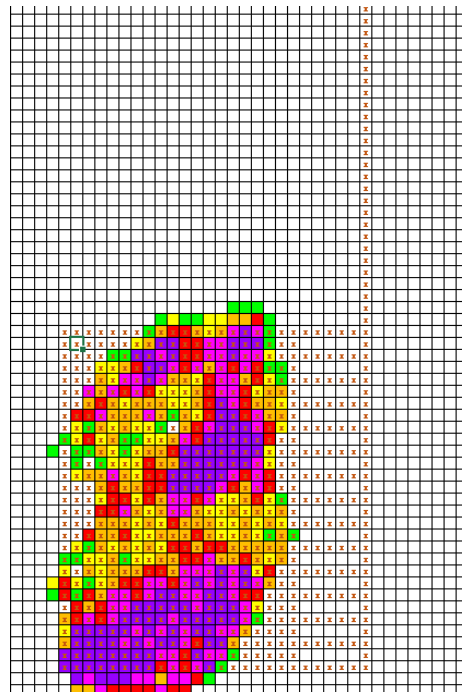
An inherent risk with Deposit 2 lies in its uncertain break-even period of 7-14 years. Although this site has better quality ore the deeper the mine, it also has a large degree of uncertainty in the rock that increases with depth. This could incur further costs relating to mitigation of environmental impacts, which include restoration of the mine site after closure, tailings storage, and impact on the community downstream. There is also potentially more equipment needed as the mine gets deeper, and additional costs are needed to maintain a level of health and safety. Thus, the project team should consider preparing extra funds moving forward to combat any uncertainties that arise and make the mining process more resilient.

Appendix A: Causal Loop Diagram of Economic, Social, and Environmental Impacts and Considerations.

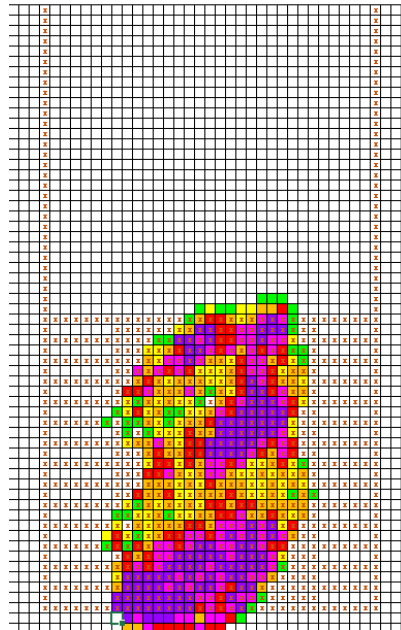


Appendix B: Block Models and Mining Data using proposed mining strategy.

Deposit 1: Block Model Proposed Mining Solution



Single Shaft - Proposed Underground Mine Plan



Dual Shaft - More Resilient Underground Mine Plan

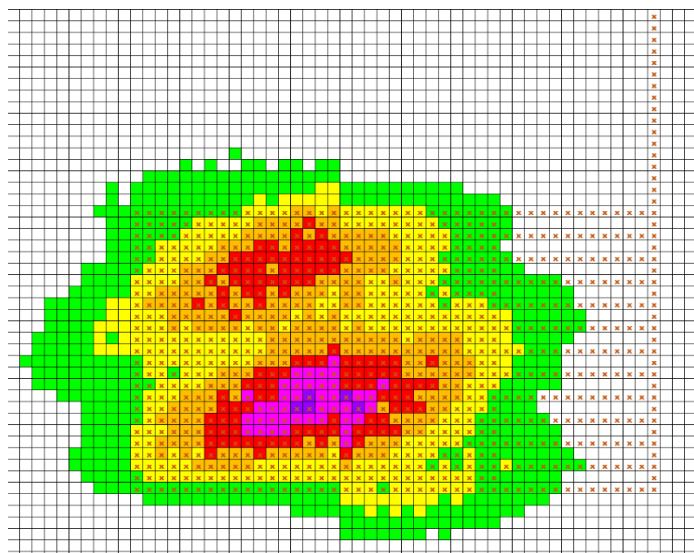
Notes:

- The cost of a second mine shaft is \$600m

OVERALL OUTPUTS	
TOTAL PROFIT	\$6,464,137,500
Waste rock	34,965,000 tonnes
Ore	63,315,000 tonnes
Copper Conc. (20%)	1,397,588 tonnes

MINE PRODUCTION LIFETIME	
UG mining rate	10,000 tonnes/day
Total time	9,828 days
	27 years

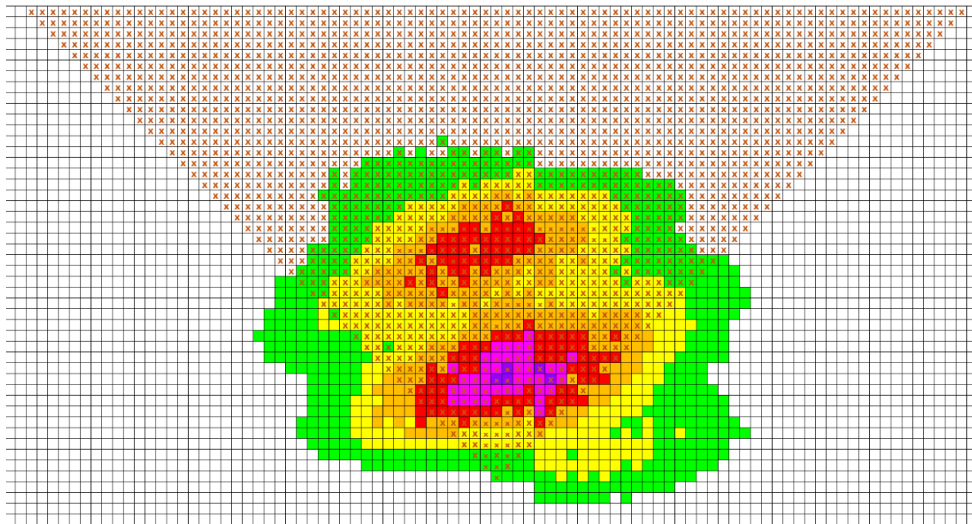
Deposit 2: Block Model Proposed Underground Mine plan



OVERALL OUTPUTS	
TOTAL PROFIT	\$8,192,643,750
Waste rock	28,957,500 tonnes
Ore	163,012,500 tonnes
Copper Conc. (20%)	2,455,819 tonnes

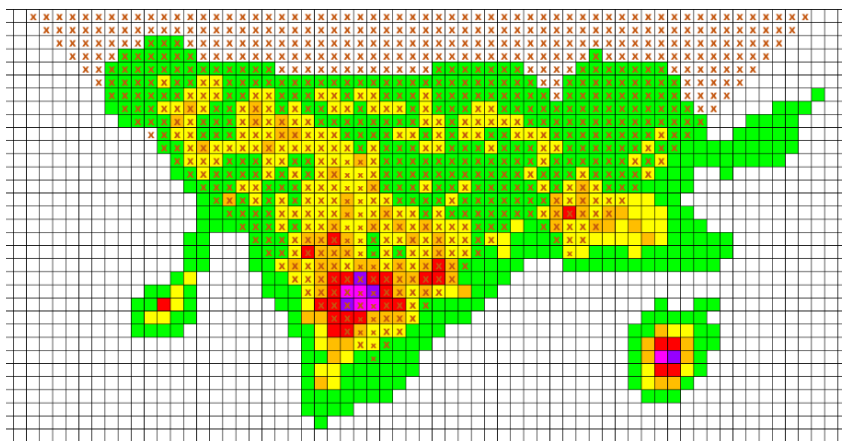
MINE PRODUCTION LIFETIME	
UG mining rate	10,000 tonnes/day
Total time	19,197 days
	53 years

Deposit 2: Block Model Proposed Open Pit Mine Plan



OVERALL OUTPUTS	
TOTAL PROFIT	\$8,259,468,750
Waste rock	245,835,000 tonnes
Ore	146,002,500 tonnes
Copper Conc. (20%)	2,164,219 tonnes

Deposit 3: Block Model Proposed Open Pit Mine Plan



OVERALL OUTPUTS	
TOTAL PROFIT	\$5,150,925,000
Waste rock	69,930,000.00 tonnes
Ore	159,570,000.00 tonnes
Copper Conc. (20%)	1,713,825 tonnes

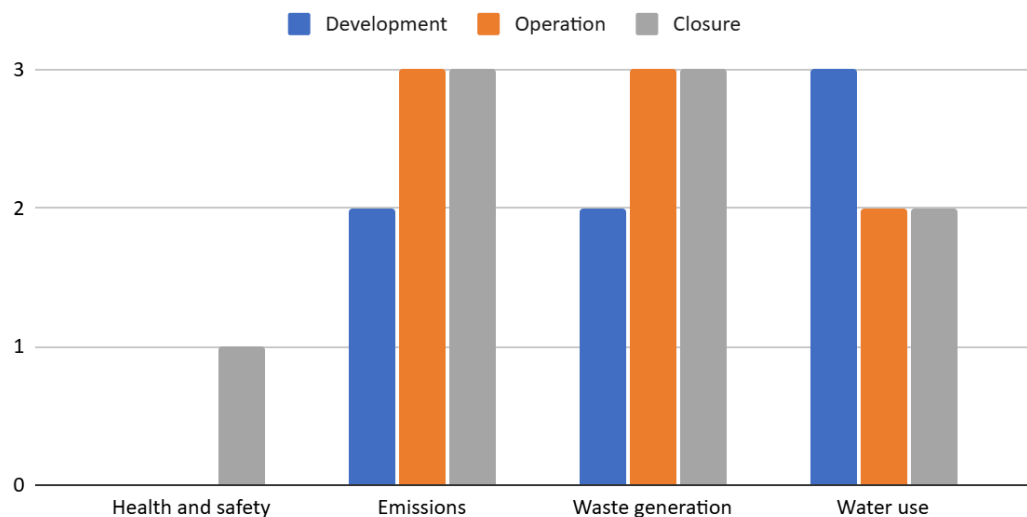
Appendix C:
Life Cycle Assessments of all three deposits.

Streamlined

Deposit 1:

	Development	Operation	Closure		Total
Health and safety	0	0	1		
Emissions	2	3	3		
Waste generation	2	3	3		
Water use	3	2	2		
Total	7	8	9		24

Development, Operation and Closure



Deposit 1 – Remote; deep, narrow; high grade

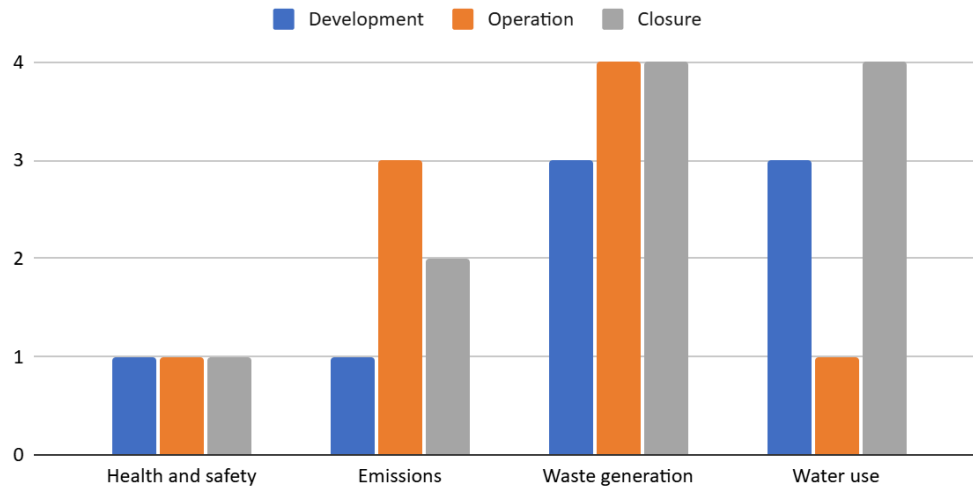
Note: In Deposit 1 the development and operation stages score a 0 in the chart for health and safety

Deposit 2:

	Development	Operation	Closure		Total
Health and safety	1	1	1		

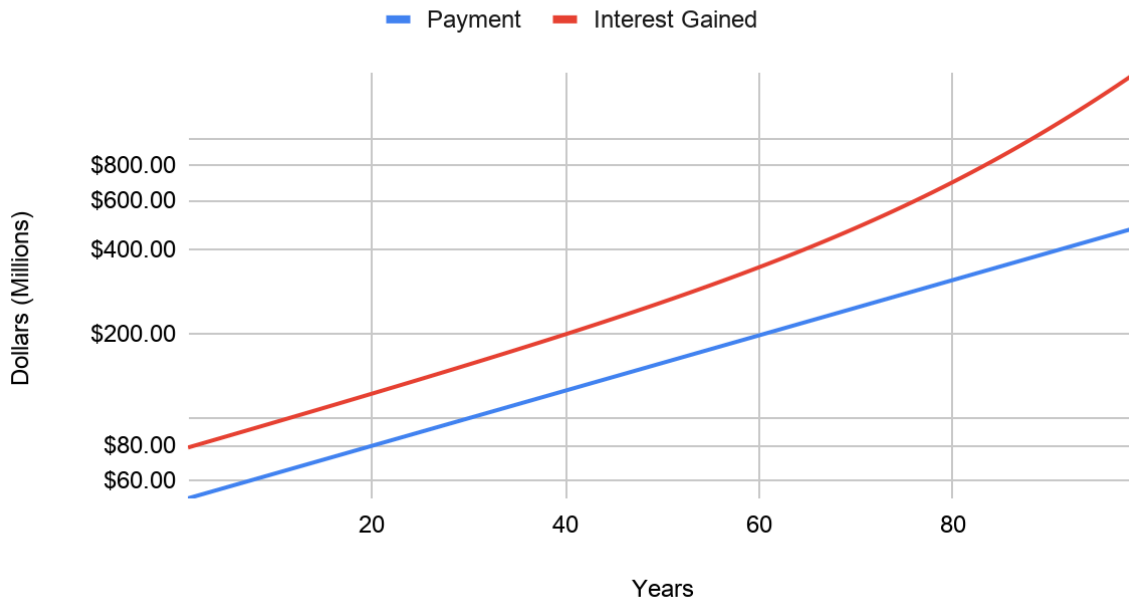
Emissions	1	3	2		
Waste generation	3	4	4		
Water use	3	1	4		
Total	8	9	11		28

Development, Operation and Closure



Deposit 2 – Moderately remote; moderately deep; grade improves with depth (UNDERGROUND)

Annual Interest Gained vs. Payment on \$970M deposit

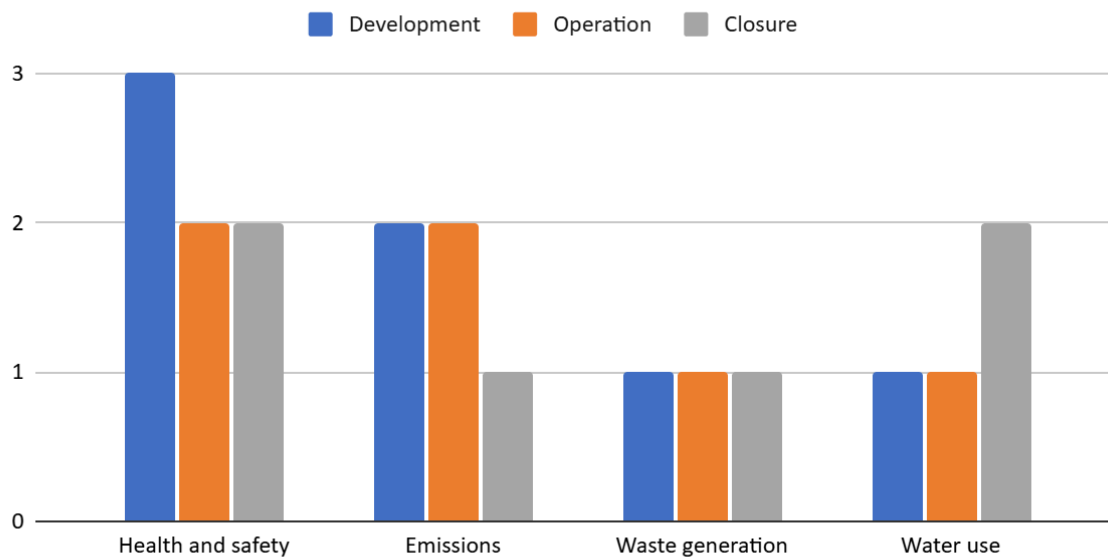


The future support fund accounts for an 8.1% annual interest rate and a 2.3% annual inflation rate. \$970M is approximately the minimum amount required so that the growth in interest outpaces inflation on a \$50M maintenance cost.

Deposit 3:

	Development	Operation	Closure		Total
Health and safety	3	2	2		
Emissions	2	2	1		
Waste generation	1	1	1		
Water use	1	1	2		
Total	7	6	6		19

Development, Operation and Closure



Deposit 3 – Near town; shallow, large; low grade

Appendix D: Weighted Decision Matrix for all three deposits.

Raw Scores:

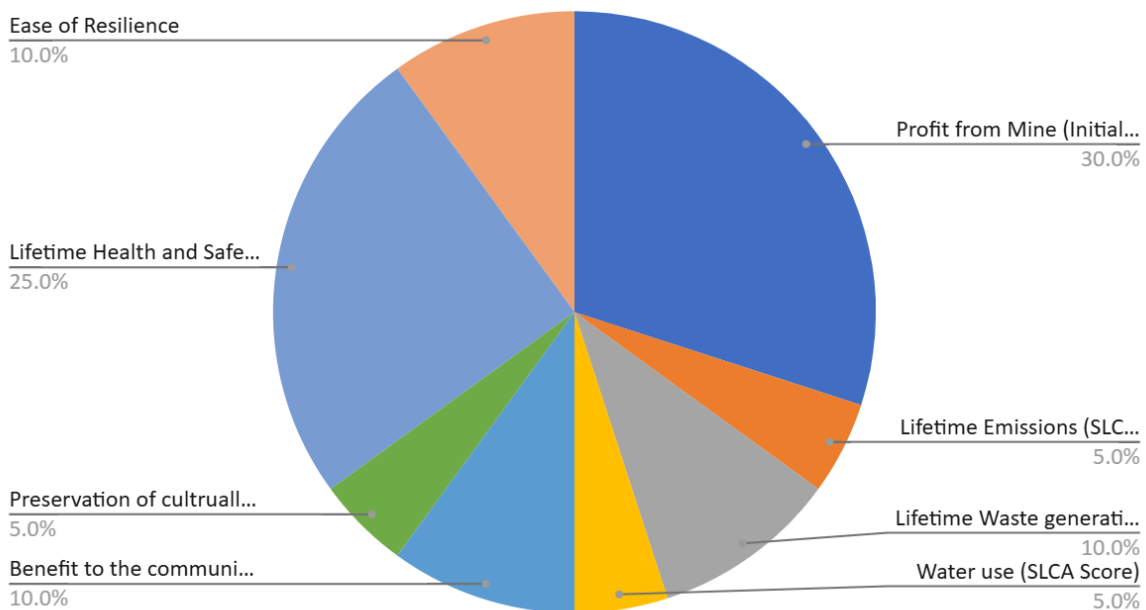
Criteria	Weighting	Deposit 1	Deposit 2	Deposit 3
Profit from Mine (Initial from Block Model, supporting infrastructure needed to be accounted)	0.3	10	10	5
Lifetime Emissions (SLCA Score)	0.05	8	6	2
Lifetime Waste generation (SLCA Score)	0.1		5	2
Water use (SLCA Score)	0.05	7	7	3
Benefit to the community (money, jobs, useful infrastructure improvements)	0.1	0.8	2.7	9.2
Preservation of culturally significant land	0.05	10	10	1
Lifetime Health and Safety (SLCA Score)	0.25	4	6	7
Ease of Resilience	0.1	4	4	6

Calculated Scores:

Profit from Mine (Initial from Block Model, supporting infrastructure needed to be accounted)		3	3	1.5
Lifetime Emissions (SLCA Score)		0.4	0.3	0.1
Lifetime Waste generation (SLCA Score)		0.7	0.5	0.2
Water use (SLCA Score)		0.35	0.35	0.15
Benefit to the community (money, jobs, useful infrastructure improvements)		0.08	0.27	0.92
Preservation of culturally significant land		0.5	0.5	0.05
Lifetime Health and Safety (SLCA Score)		1	1.5	1.75
Ease of Resilience		0.4	0.4	0.6

Total		6.43	6.82	5.27
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Weighting vs. Criteria



Appendix E:**Deposit 1 Infrastructure Cost Rough Estimate Distribution**

Component	Estimated Share of Total Infrastructure Cost	Estimated % of Mining Profits Lost
Road Construction (300km)	45%	15-25%
Bridge Over Nicus River	10%	3-8%
Power Transmission (Over 300km)	35%	12-20%
Environmental Preservation/ Monitoring	10%	3-6%
Total		33-59%

Appendix F:

Map of Mining Deposits Source: (Module 6 - Case Study Guidelines-2025W.pdf)



Statement of GenAI use: GenAI was used to verify the grammar and spelling of this letter. All ideas and content were produced without GenAI assistance.

Works Cited

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